



PIMPRI CHINCHWAD EDUCATION TRUST'S.  
**PIMPRI CHINCHWAD COLLEGE OF ENGINEERING**  
(An Autonomous Institute)  
Affiliated to Savitribai Phule Pune University (SPPU)  
ISO 21001:2018 Certified by TUV SUD

**T.Y. B. TECH**

**Year: 2025 – 26**

**Subject: FSDL**

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**Branch:** Computer Engineering

**Division: C**

## Assignment No 4

**Title : Create any interactive data visualization dashboard for any real-life application like weather dashboard. Visualize data using charts and graphs (use libraries like Chart.js or D3.js).**

### Objectives:

- To understand the structure of a dynamic data visualization dashboard
- To learn how HTML is used to create layout and dashboard components
- To apply CSS and Bootstrap for styling and responsive UI
- To use JavaScript for fetching and visualizing real-time data
- To implement charts and graphs using Chart.js
- To design an interactive and user-friendly weather dashboard

### Outcomes:

- Design a functional weather dashboard interface
- Display weather data dynamically using charts and cards
- Apply responsive design using Bootstrap CDN
- Implement interactivity using JavaScript and APIs
- Improve understanding of front-end development and data visualization

### Theory :

A Student Performance Dashboard is a web-based data visualization system used to display and analyze students' academic performance in a graphical format.

Instead of showing marks only in tables, a dashboard uses charts and graphs to make data easy to understand. Visual representation helps teachers and students quickly analyze performance trends, compare subjects, and identify areas of improvement.

In educational institutions, large amounts of academic data are generated such as:

- Subject-wise marks
- Attendance percentage
- Semester-wise results
- Overall grades
- Performance trends over time

Manually analyzing this data can be difficult. Therefore, a dashboard system helps in converting raw data into meaningful visual insights.

## Use of Charts in Dashboard

Charts are important in understanding student performance patterns:

- **Bar Chart** – Shows subject-wise marks comparison
- **Line Chart** – Shows performance trend across semesters
- **Pie Chart** – Displays grade or attendance distribution

### Code :

```
<!DOCTYPE html>

<html lang="en">

<head>

  <meta charset="UTF-8">

  <title>Student Performance Dashboard</title>

  <link rel="stylesheet" href="style.css">

  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

</head>

<body>

  <h1>Student Performance Dashboard</h1>

  <div class="controls">

    <input type="text" id="searchInput" placeholder="Search student">

    <select id="studentSelect"></select>

  </div>

  <div class="info-bar">

    <div id="nameBox">Name:</div>

    <div id="attendanceBox">Attendance:</div>

    <div id="statusBox">Status:</div>

  </div>
```

```
<div class="dashboard">
```

```
<div class="card">
```

```
<h3>Semester Progress</h3>
```

```
<canvas id="progressChart"></canvas>
```

```
</div>
```

```
<div class="card">
```

```
<h3>Pass vs Fail</h3>
```

```
<canvas id="passFailChart"></canvas>
```

```
</div>
```

```
<div class="card full-width">
```

```
<h3>Top 5 Students (Sem 4)</h3>
```

```
<canvas id="topStudentsChart"></canvas>
```

```
</div>
```

```
<div class="card full-width">
```

```
<h3>Attendance Trend</h3>
```

```
<canvas id="attendanceChart"></canvas>
```

```
</div>
```

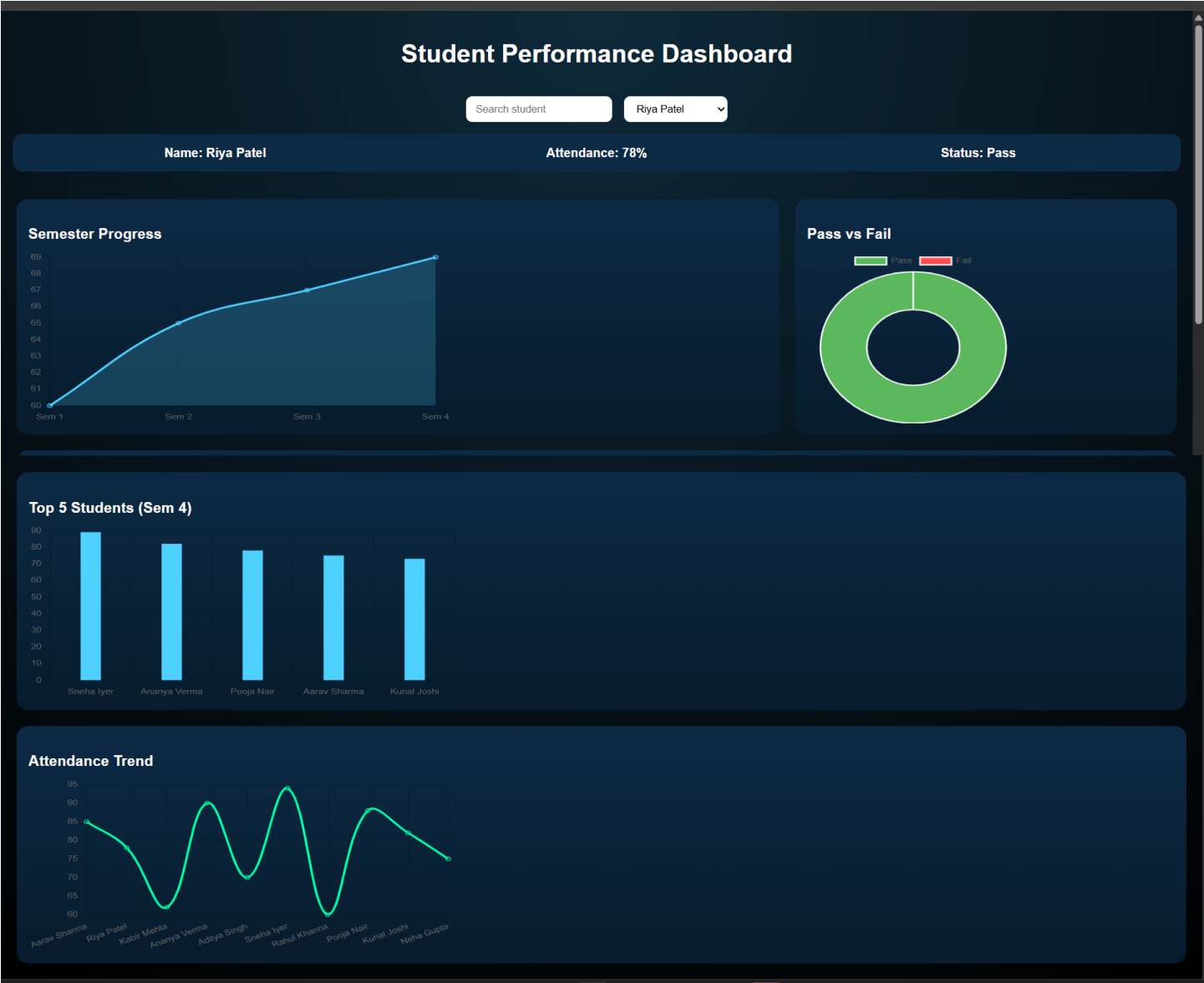
```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

Output :



**Conclusion :**

The Student Performance Dashboard successfully demonstrates how academic data can be converted into visual charts for better understanding. It helps in analyzing subject-wise marks, attendance, and overall progress easily.

This project improved our knowledge of data visualization and front-end development using Chart.js and Bootstrap.